

Integrated Framework for Surgery Scheduling Management under Uncertainty: From Strategic to Tactical Planning

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Abstract

Abstract text.

Keywords:

1. Related Works

Surgery scheduling problems have traditionally been classified into three planning levels with different sources of uncertainty: strategic, tactical, and operational (Rahimi and Gandomi, 2020; Cardoen et al., 2010; Wang et al., 2021; Melo and Fogliatto, 2025). At the strategic level, long-term decisions define the desired split of the capacity into specific surgical specialities and specify patient volumes of different medical specialties to be admitted over the planning horizon (Burdett et al., 2023) — a process commonly referred to as Case Mix Planning (CMP). Different objectives have been introduced at this level, such as maximizing profit (Fügner, 2015; Ma and Demeulemeester, 2013) or minimizing long-term costs (Hof et al., 2015; McRae and Brunner, 2020). The tactical level defines a cyclic schedule (often weekly) commonly known as Master Surgery Schedule (MSS) (Bovim et al., 2020; Siqueira et al., 2018; Carneiro et al., 2025). The MSS step allocates time blocks in operating theatres to surgical teams and is reviewed in a medium-term horizon (e.g., monthly or quarterly). Finally, the operational level focuses on the daily operation and derives a Surgical Case Schedule (SCS), which generally assigns individual patients to time blocks previously allocated to specific surgical teams at the tactical level (Vieira et al., 2025; Harris and Claudio, 2022).

Despite the large body of literature in surgery scheduling, there are significant gaps to be bridged. For example, inter-level integration remains limited in scope, often focusing on short-term alignment between surgeries and post-surgical bed occupation, and generally relying on deterministic approaches (Melo and Fogliatto, 2025; Wang et al., 2021; Siqueira et al., 2018). Furthermore, regardless of the planning level, strategies with a focus on guaranteeing waiting time requirements for patients still remain to be proposed. This is probably due to the complex propagation of uncertainty over different planning levels, which demands hybrid approaches that combine mathematical programming, stochastic modelling and queuing theory.

The inherent interconnection between the strategic and tactical levels has motivated unified frameworks that link CMP and MSS decisions (Melo and Fogliatto, 2025). These often adopt sequential or joint optimisation approaches where case mix decisions act as the input to surgery scheduling models (Ma and Demeulemeester, 2013; Fügner, 2015). Modelling approaches typically consist of Mixed Integer Programming (MIP) formulations that focus on maximising hospital profit subject to Operating Theatre (OT) and bed capacity constraints. While these frameworks acknowledge uncertainty, it is generally confined to downstream elements such as length of stay or ward occupancy. Although modelling post-surgical congestion is relevant, such partial treatments remain short-sighted when not complemented by uncertainty in surgery durations and, crucially, by an explicit modelling of dynamic waiting lists across surgical specialities, which constitute the starting point of the entire planning process.

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The perspective can be broadened by incorporating additional uncertainty sources and dynamic elements. For instance, Liu et al. (2019) model elective admissions through a Markov Decision Process (MDP), highlighting the role of waiting-list size in guiding strategic decisions. Despite its conceptual value, the framework disregards essential practical constraints, such as surgeon availability, block scheduling and case-mix heterogeneity, thereby limiting practical applicability. Similarly, McRae and Brunner (2020) demonstrate that explicitly modelling uncertainty and selecting appropriate aggregation levels can significantly improve CMP outcomes and enable joint CMP-MSS optimisation. Nevertheless, waiting-list dynamics remain outside of the modelling scope, implying no full integration of strategic and tactical decisions. Furthermore, their reliance on Sample Average Approximation (SAA) leads to representations that may understate distributional variability. These studies underscore the value of integrating CMP and MSS, yet they also reveal persistent shortcomings: uncertainty is represented only selectively across planning levels, waiting-list requirements are rarely modelled explicitly, and key operational constraints required for real-world implementation remain overlooked.

Uncertainty in surgery duration is another important issue that, when disregarded, may result in unpredictable idle or overtime in OTs (Shehadeh and Padman, 2022). Whilst overtime is costly and can lead to cancellations that compromise the quality of care, idle time reflects an inadequate use of OT capacity (Shehadeh and Padman, 2022; Harris and Claudio, 2022; Kayvanfar et al., 2025). With the aim of maximising hospital revenue while preventing excessive overtime, Barz and Rajaram (2015) propose a tactical patient admission and scheduling problem that derives a CMP under multiple resource constraints and stochastic clinical progression. The approach utilises a newsvendor heuristic to reserve capacity for emergency arrivals, and an MDP model to select the patient cohort to be admitted whilst considering probabilistic transitions between clinical states. The model, however, excludes the possibility of cancellations or postponements driven by overtime, as it assumes that the hospital always disposes of the resources to treat each admitted patient. Moreover, by treating resource consumption as deterministic, the framework effectively ignores uncertainty in surgery duration — one of the main drivers of operational instability. As a result, the model underestimates the practical interplay between stochastic surgery durations, cancellations and overtime, limiting its applicability to integrated CMP-MSS decision-making.

Furthermore, Zhang et al. (2020) study a tactical surgery planning with the explicit aim of mitigating overtime risk when allocating elective patients to OTs over an upcoming planning horizon. Their model accounts for stochastic surgery durations and adopts a risk-averse objective that jointly captures both the probability and the expected magnitude of overtime, extending earlier approaches that consider only one of these dimensions (Shylo et al., 2013; Rath et al., 2017). Although they derive the durations from real-world data, eventually they approximate it through a set of discrete scenarios, limiting the full distributional representation. A similar treatment appears in Shehadeh et al. (2026), which introduces a Distributionally Robust Optimisation (DRO) approach to address uncertainty in surgery durations. While their approach minimises worst-case expected cost over an ambiguity set defined by known moments and support, it likewise relies on scenario-based representations. Accurately capturing surgery-duration variability requires a large number of scenarios, substantially increasing computational burden and potentially undermining the practical advantages of DRO and related methods.

These limitations regarding the integration of overtime management and its associated cancellation risks into surgical planning frameworks highlight important research gaps. There remains a need for models that explicitly incorporate cancellation dynamics induced by overtime and that provide a more thorough representation of surgery-duration uncertainty beyond scenario-based approximations. A key challenge which we tackle in this paper is to develop a modular approach to capture full distributional characteristics to build a planning model and use its output to feed an integrated CMP-MSS approach — i.e., not solving an entire model at once.

An additional and often underexplored dimension in integrated surgical planning concerns the incorporation of practical constraints inherent to real-world healthcare settings. Certain surgical activities, such as organ transplantation, impose rigid scheduling requirements that are difficult to reconcile with standard optimisation assumptions (da Silva et al., 2025). Furthermore, public hospitals in developing countries (e.g., Brazil) operate under severe resource constraints, high and volatile demand, and heterogeneous case mixes, all of which exacerbate the complexity of surgical planning (Carneiro et al., 2025; Siqueira et al., 2018; da Silva et al., 2025). These contextual factors challenge the scalability and transferability of many optimisation-based approaches, which often presume stable resources, predictable demand and homogeneous patient populations. Indeed, as noted by Melo and Fogliatto (2025), much of the existing literature is calibrated to relatively small hospitals in developed regions, limiting its relevance for large, resource-constrained systems. Ignoring such operational realities risks producing models that are theoretically stable

but difficult to implement in practice, underscoring the need for scheduling frameworks that explicitly accommodate institutional, organisational and contextual constraints.

This paper addresses the identified gaps in the surgical planning literature by proposing an integrated framework that jointly links strategic and tactical decision levels while explicitly accounting for key sources of uncertainty. The framework captures dynamic surgical demand through explicit waiting-list modelling, enabling the enforcement of waiting-time requirements and long-term queue stability (Shortle et al., 2018), and incorporates stochastic surgery durations to properly balance cancellation, idle time, and overtime risks for each surgical session. The proposed approach is grounded in a real-world application at the Clinical Hospital of the University of Campinas (HC-Unicamp) in Brazil, a high-complexity public hospital that provides detailed operational data and a challenging planning environment (HC - Unicamp, 2024b).

The main contributions of this work are as follows:

- A stochastic waiting-list management strategy based on an (R, Q) control policy (Axsäter, 2015) that ensures compliance with waiting-time targets while maintaining long-term queue stability across surgical specialties;
- A cancellation risk management mechanism based on a newsvendor formulation (Arrow et al., 1951), which determines optimal time buffers for the last surgery in each session under duration uncertainty and defines probability distributions for the number of planned surgeries that can be accommodated without incurring overtime;
- A MIP model for operating theatre allocation that: (i) incorporates practical constraints, including those associated with transplant surgeries, while remaining computationally tractable; and (ii) explicitly incorporates the waiting-list management strategy and cancellation risk considerations from the previous items;
- An empirical validation of the integrated framework with data from a large, resource-constrained public hospital, demonstrating its operational relevance and implementability.

By jointly addressing relevant sources of uncertainty within a unified framework, this work introduces an innovative and integrated surgical planning approach that advances the literature by bridging multiple relevant gaps. The approach also bridges the gap between theory and practice by bringing forward a practically grounded and empirically validated framework for long-term management of surgery queues within complex healthcare settings.

2. Problem Description

Hospital de Clínicas Unicamp (HC - Unicamp) is a university hospital located in Campinas, Brazil (HC - Unicamp, 2024b). It is a high-complexity public hospital serving approximately 6.5 million inhabitants from Campinas' metropolitan area and other regions of Brazil that handles around 500,000 patient cases per year. Of these, roughly 10,000 correspond to elective surgical procedures, with the remainder involving outpatient consultations and emergency care (HC - Unicamp, 2024a). In addition to providing healthcare services, HC - Unicamp operates as a teaching and research hospital, training healthcare professionals and conducting medical and interdisciplinary research.

Elective surgical patients follow a complex, multi-stage care pathway (see Figure 1), encompassing pre-operative, intra-operative and post-operative phases. This process begins with waiting-list management, initiated at local medical consultations and coordinated through the regional bed regulation centre (HC - Unicamp, 2024c), a government agency responsible for regulating hospital access across Campinas Metropolitan Region. Patient entry into HC - Unicamp typically occurs via scheduled outpatient appointments, although some emergency cases subsequently become elective and are incorporated into the same planning process. Once referred, patients are assigned to waiting lists according to medical subspecialty. Decisions at this stage directly determine the CMP (i.e., strategic layer), influencing waiting times, speciality-level demand balance and the long-term utilisation of operating theatre capacity (Siqueira et al., 2018).

On the day of surgery, patients enter the upstream unit management phase, where clinical reassessment is performed. Based on clinical requirements, Intensive Care Unit (ICU) support may be required pre- or postoperatively, necessitating prior bed reservation; otherwise, postoperative ward bed availability must be ensured. Patients then undergo pre-anaesthetic evaluation, diagnostic tests and surgical safety checks. If contraindications arise, the procedure is suspended and the patient is either discharged or rescheduled, according to clinical severity. If cleared, the patient

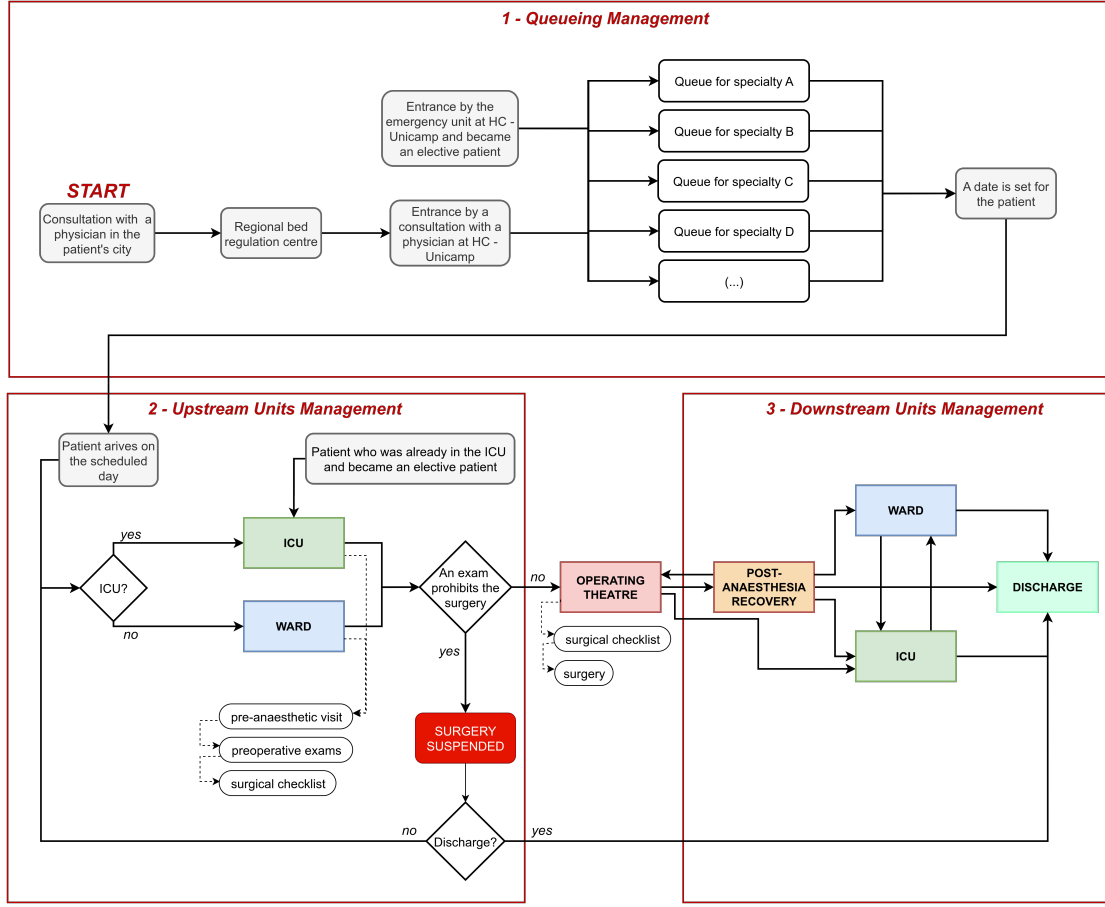


Figure 1: Patient flow at HC - Unicamp.

proceeds to the operating theatre, where the surgical procedure and subsequent theatre cleaning are performed. In the present study, ward and ICU capacity are sufficient and therefore do not constitute binding constraints, allowing the analysis to focus on waiting-list management, uncertainty in surgery duration and operating theatre allocation.

Once in the operating theatre, the patient is managed by a dedicated surgical team and anaesthetist. This stage includes final safety checks, the surgical procedure itself and post-operative cleaning. Here, uncertainty in surgery duration plays a central role, as it directly affects idle time, overtime, and the risk of cancellations. Indeed, it is well known that surgery duration variability has a significant impact on operating theatre utilisation and overall system throughput (e.g., Pandit and Tavare, 2011), therefore meriting explicit consideration in the proposed framework.

Following surgery, patients are transferred to the Post-Anaesthesia Recovery Unit (PARU), with direct ICU transfer occurring only in exceptional cases. After stabilisation, patients are admitted to a ward or ICU bed and subsequently discharged. HC - Unicamp has 15 operating theatres dedicated to elective surgeries: 13 located in the Elective Surgery Centre (ESC) and 2 in the Ambulatory Surgery Centre (ASC), which operates exclusively during morning shifts. These theatres are heterogeneous with respect to infrastructure and equipment, leading to compatibility restrictions across medical subspecialties.

The planning horizon comprises approximately 20 working days per month, each divided into two 6-hour shifts (morning and afternoon). Henceforth, we refer to a theatre shift as a *block*. Hospital policy assigns each block to a single subspecialty, while allowing the same subspecialty to occupy multiple blocks within the same theatre, provided compatibility conditions are met. Sequential allocation is therefore permitted (and preferred by hospital managers), enabling a subspecialty assigned to a morning block to also occupy the corresponding afternoon block.

Each month, operating theatres must be allocated across 40 medical subspecialties. Relevant subsets include: (i) those that mandatorily require two consecutive blocks in the same theatre due to the extended duration of their procedures; (ii) priority subspecialties, which take precedence in allocation decisions in accordance with institutional rules; and (iii) transplant subspecialties, which impose additional operational constraints.

Block allocation feasibility depends mainly on the availability of surgical teams and anaesthetists. A subspecialty can only be assigned to a block if an appropriate team is available on the corresponding day and shift. Moreover, the number of simultaneous allocations requiring anaesthesia cannot exceed the number of anaesthetists available in each period. For example, if 15 anaesthetists are available on a given morning, at most 15 anaesthesia-dependent procedures may be scheduled concurrently.

Day	PERIOD/THEATRE	1	2	3	4	5	6	7	8	9	10	11	12	17	ASC-1	ASC-2
1	MORNING	card sur	tumour orth		procto	uro onco	anaest dor		gast liv	otorhi			vasc			
	AFTERNOON	card sur	tumour orth	pc local	procto	uro onco				otorhi	gast gdp/vb		vasc	vasc local		
2	MORNING	card sur	foot orth	plast	trauma orth	uro tx		uro tx	gast eed	otorhi	gast eed	neuro	trauma orth			
	AFTERNOON	card sur		plast						otorhi	gast eed	neuro	trauma orth			
3	MORNING	card sur	knee orth	plast	thorax sur	vasc local	paed sur	plast		otorhi	head neck	neuro	trauma orth	vasc		
	AFTERNOON	card sur		plast	thorax sur		paed sur	plast		otorhi	gast obes	neuro		vasc		
4	MORNING	card sur	eef	adult endo	gast gdp/vb	uro lithiasi	vasc	adult endo	gast liv	otorhi	gast obes	neuro	neuro colu		ophtal	
	AFTERNOON	card sur	pc local	ophtal	gast gdp/vb	uro lithiasi	vasc	otorhi	gast liv	otorhi	gast obes	neuro	neuro colu			
5	MORNING		paed orth	plast		uro gynec	paed sur			head neck	head neck	neuro colu			hand orth	
	AFTERNOON		hip orth										trauma orth			

Figure 2: Excerpt from a week’s operating theatre schedule.

Figure 2 presents an excerpt from a monthly operating theatre schedule. Rows represent day and shift, and columns correspond to theatres; allocated blocks are colour-coded (blue for procedures requiring anaesthesia and grey otherwise). Blank blocks highlight unallocated slots, reflecting the complexity of the constraint set. Allocations shown in red correspond to transplant surgeries, which require the simultaneous allocation of two theatres within the same day and shift.

Historically, schedules such as those shown in Figure 2 have been manually constructed using only electronic spreadsheets, often requiring more than two full working days. This manual process is time-consuming and does not guarantee optimality with respect to institutional policies. Automating the scheduling process therefore reduces administrative burden and enables the generation of optimal solutions under complex constraints.

In addition to the impact of automating surgery scheduling in a reference hospital in Brazil, this paper provides an approach that is firmly grounded in reality and is therefore adaptable to other general hospitals. By jointly managing waiting-list dynamics, surgery duration uncertainty and operating theatre allocation, our proposed approach provides a coherent and practically grounded solution capable of satisfying waiting-time requirements while mitigating idle time, overtime and cancellation risks.

3. Proposed Framework

We propose an integrated framework for elective surgery planning under uncertainty that links waiting-list management, in-block time management, and operating theatre scheduling. Rather than formulating these decisions as a single monolithic model, the framework decomposes the problem into three modules. The first two modules address strategic decisions related to queue and overtime management, and their outputs are then used as inputs to the tactical operating theatre scheduling model. Figure 3 summarises the proposed framework.

The first module addresses queue management by determining admission targets based on demand characteristics and waiting-time requirements. Specifically, a modified (R, Q) inventory policy uses the probability distribution of patient demand for each medical subspecialty to define lower and upper bounds on the number of patients to be

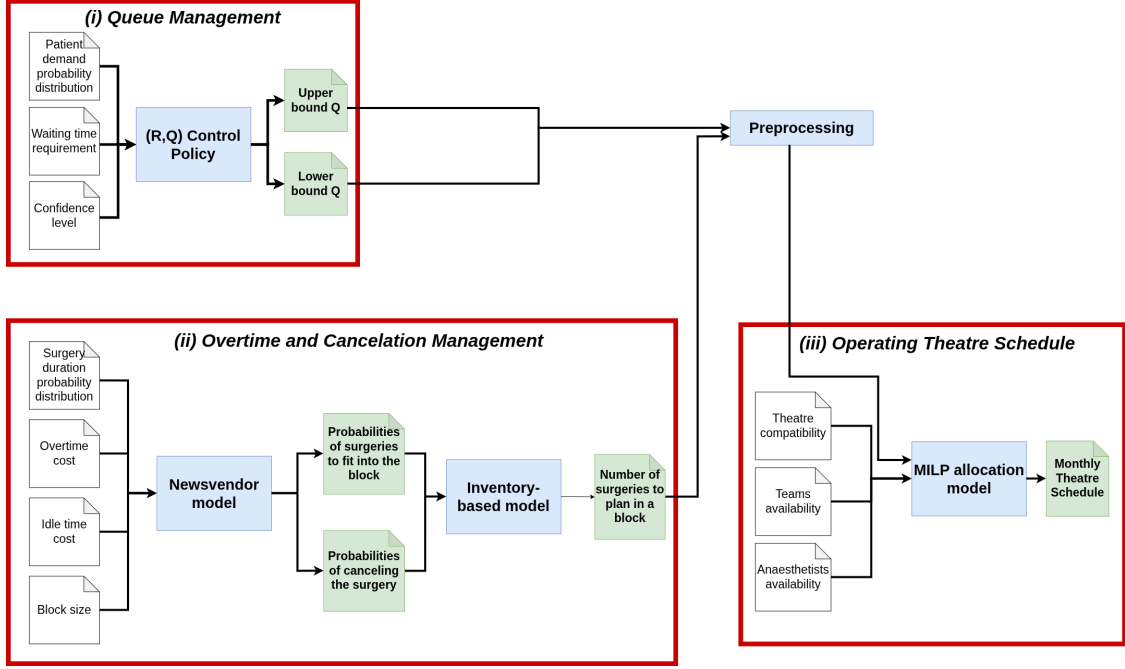


Figure 3: Proposed framework.

admitted each month, while ensuring that waiting-time requirements are satisfied at a given confidence level. This module supports the long-term balance of the waiting list by aligning admission decisions with demand uncertainty and institutional access targets.

The second module addresses overtime and cancellation management by evaluating the trade-off between overtime, idle time, and the risk of procedure cancellation under stochastic surgery durations. Essentially, a Newsvendor-based model is used to determine an optimal time for the allocation of the final surgery in a block, which is then translated into a target number of surgeries to be planned in each block, balancing the costs of overtime and idle time against the risk of exceeding block capacity.

The outputs of these two modules provide key inputs to the third module, which formulates the OT scheduling problem. In this way, monthly theatre allocation decisions are informed not only by resource availability and clinical compatibility, but also by demand uncertainty and block-level execution risk. At the same time, the scheduling model remains deterministic, which facilitates solution within practical computational times for real-world implementation.

Overall, the proposed framework provides a coherent approach to elective surgery planning across multiple decision levels. By explicitly incorporating uncertainty into the strategic planning stages, it supports more robust and operationally meaningful scheduling decisions. The following subsections present each module in detail. To help understanding the present study, we introduce the acronyms used in this paper in Table 1. For the mathematical notation, we refer the reader to the tables in each module's section.

4. Long-Term Waiting List Management

As discussed in Section 1, CMP constitutes the strategic decision layer responsible for determining the volumes of elective patients to be admitted over a given planning horizon. In contrast to much of the existing literature, which primarily selects these volumes to optimise financial or capacity-related objectives, this work adopts a stability-oriented perspective. We aim to enable decision-makers to impose maximum acceptable waiting times (e.g., six months) and to guarantee, in the long run, that such targets are met without allowing waiting lists to grow unboundedly. This is particularly relevant in the context of public healthcare systems, such as Brazil's Unified Health System (SUS),

Table 1: Acronyms used in the paper.

Acronym	Description
CMP	Case Mix Planning
MSS	Master Surgery Scheduling
SCS	Surgical Case Scheduling
OT	Operating Theatre
MDP	Markov Decision Process
SUS	Brazilian Unified Health System
HC - Unicamp	Hospital de Clínicas Unicamp
OTSP	Operating Theatre Scheduling Problem
MILP	Mixed-Integer Linear Programming
PMF	Probability Mass Function
CDF	Cumulative Distribution Function

where there is a legal obligation to provide care to all patients in need. Thus, ensuring equitable access to care and adherence to waiting time standards is paramount.

In line with this objective, this section introduces the theoretical foundations of the modified (R, Q) control policy proposed to manage elective surgery waiting lists under stochastic demand. We first characterise the dynamics of the underlying stochastic process and analyse its long-run behaviour. Subsequently, we derive analytical upper and lower bounds on the batch size Q that ensures compliance with prescribed waiting-time requirements, thereby establishing the strategic input for the integrated planning framework. Table 2 summarises the notation used in this section.

4.1. (R, Q) surgery scheduling policy: long-term distributions

Let X_t , $t \geq 0$ be a stochastic process representing an elective surgery queue with stationary and independent increments over discrete planning periods, with X_t representing the number of patients in the system at period $t \geq 0$. Assume that the queue is controlled via a modified (R, Q) policy (e.g., Axsäter, 2015) whereby a batch of $Q \in \mathbb{N}$ surgeries is processed at time period t whenever $X_t \geq R$, $R \geq Q$ at the start of the period, where $\mathbb{N}$ is the set of positive integers. In contrast to the classical (R, Q) policy, which processes as many batches as necessary to bring the inventory below the reorder point (R), our policy can only process a batch at a time and relies on long-term properties to ensure that the process X_t , $t \geq 0$ is stable and guarantees prescribed waiting time requirements.

Let D be a non-negative integer random variable representing the overall demand between two consecutive planning periods, with probability distribution $f_D : \Omega \rightarrow [0, 1)$, $\Omega \subset \mathbb{Z}_+$, and cumulative distribution $F_D : \mathbb{Z}_+ \rightarrow [0, 1]$ given by

$$F_D(d) = \sum_{l=0}^d f_D(l),$$

where $\mathbb{Z}_+$ denotes the set of non-negative integers. This implies that the dynamics of process X_t , $t \geq 0$ can be expressed as:

$$X_{t+1} = X_t + d - Q \mathbb{1}_{\{X_t \geq R\}}, \quad (1)$$

where $\mathbb{1}_{\{A\}}$ is the indicator function, which is equal to 1 if A holds and is nil otherwise, and d is a value drawn from f_D that represents the surgical demand in period t . The state space of process X_t , $t \geq 0$ is given by $S = \{0, 1, 2, \dots\}$, where $X_t = i$ indicates that there are $R - Q + i$ patients waiting for elective surgery.

Letting

$$Y_t = \left\lfloor \frac{X_t}{Q} \right\rfloor, \quad Z_t = X_t \% Q, \quad (2)$$

where $\%$ represents the modulus operator, we can write:

$$X_t = QY_t + Z_t,$$

where Y_t , $t \geq 0$ is the integer number of batches required to bring the queue instantaneously below R and $Z_t \in \{0, 1, \dots, Q-1\}$ is the number of patients that would remain in excess of the minimum allowed queue of $(R - Q)$ patients after removing Y_t batches of Q patients instantaneously.

Note that, from the definitions above, the state space of process Y_t , $t \geq 0$ is $S_Y = \mathbb{Z}_+$, while $S_Z = \{0, 1, \dots, Q-1\}$ is the state space of process Z_t , $t \geq 0$. In the next result, we will derive the long-term distribution of process Z_t , $t \geq 0$.

Table 2: Symbols used in the long-term waiting list management model.

Index	Description
t	Planning period index
z	State index of the remainder process
i, j	Generic state indices
Set	Description
S	State space of process X_t
S_Y	State space of process Y_t
S_Z	State space of process Z_t
Ω	Support of the demand random variable
Parameter	Description
R	Reorder threshold of the modified (R, Q) policy
Q	Batch size under the modified (R, Q) policy
τ	Waiting-time target
α	Probability threshold for the waiting-time requirement
$\bar{Q}$	Upper bound on feasible batch sizes
$\underline{Q}$	Lower bound on feasible batch sizes
M	Maximum number of batch arrivals in one period
Random Variable / Stochastic Process	Description
X_t	Queue-length process
D	Demand between two consecutive planning periods
d	Realisation of D
Y_t	Batch-level queue process
Z_t	Remainder process associated with X_t
A	Number of batch arrivals in one period
ω	Waiting time of an arriving patient
Function	Description
$f_D(\cdot)$	Probability mass function of D
$F_D(\cdot)$	Cumulative distribution function of D
$A(z)$	Number of batch arrivals given remainder state z
$\pi_Z(\cdot)$	Steady-state distribution of Z_t
$\pi_Y(\cdot)$	Steady-state distribution of Y_t
P_Z	Transition matrix of process Z_t
P^Y	Transition matrix of process Y_t
$p_{zz'}^Z(k)$	Conditional transition probability of Z_t given demand k
p_{ij}	Generic transition probability entry
Derived Quantity	Description
$\mathbb{1}_{\{X_t \geq R\}}$	Indicator of batch processing

Proposition 1. Let $\pi_Z : S_Z \rightarrow [0, 1]$ be the steady state distribution of process X_t , $t \geq 0$. Then, it holds that

$$\pi_Z(z) = \frac{1}{Q}, \forall z \in S_Z. \quad (3)$$

Proof. The proof closely follows that of Axsäter (2015, Proposition 5.1). Let P_Z be the transition matrix describing the dynamics of process Z_t , $t \geq 0$. Assume that $Z_t = z$, $z \in S_Z$ at the start of period $t \geq 0$, and let $D = k$ be the number of new patient arrivals in period t . Hence, it follows that

$$Z_{t+1} = (z + k) \% Q = (z + k + QY_t) \% Q = (X_t + k) \% Q,$$

where the last equality come from the definitions in Eq. (2). Therefore, given $D = k$, it is evident that $p_{zz'}^Z(k) := P(Z_{t+1} = z' | Z_t = z, D = k)$ equals one for exactly one value of z and zero otherwise. Consequently, we can write:

$$\sum_{z=0}^{Q-1} p_{zz'}^Z = \sum_{z=0}^{Q-1} \mathbb{E}_k \{p_{zz'}^Z(k)\} = \mathbb{E}_k \left\{ \sum_{z=0}^{Q-1} p_{zz'}^Z(k) \right\} = \mathbb{E}_k \{1\} = 1.$$

The expression above implies that the Markov chain describing the dynamics of process Z_t , $t \geq 0$ is doubly stochastic,

and therefore has a uniform steady state distribution. Indeed, substituting (3) in the steady state equations:

$$\pi^Z(z) = \sum_{z'=0}^{Q-1} \pi^Z(z') p_{z'z}^Z,$$

one can easily see that the steady state distribution satisfies Eq. (3). $\square$

The result in Proposition 1 is very important, as it shows that the steady state distribution of process Z_t , $t \geq 0$, is uniform regardless of the distribution of patient arrivals. Next, we will use this fact to derive the steady state distribution of process Y_t , $t \geq 0$.

4.2. The dynamics of process Y_t , $t \geq 0$

Assume that process Z_t , $t \geq 0$ has reached equilibrium, which is also equivalent to supposing that $P(Z_0 = z) = \pi_Z(z) = \frac{1}{Q}$ (e.g., Brémaud, 2020). We can describe the dynamics of process Y_t , $t \geq 0$ as:

$$Y_{t+1} = \begin{cases} Y_t - 1 + A, & \text{if } Y_t \geq 0, \\ A; & \text{if } Y_t = 0, \end{cases} \quad (4)$$

where A is the round number of batch arrivals at time t . To fully characterize the random variable A , it is necessary to define

$$A(z) = \left\lfloor \frac{D+z}{Q} \right\rfloor, \quad (5)$$

which represents the round number of batch arrivals at time t , given that $Z_t = z$. Then, in view of Proposition 1, it follows that:

$$P(A = k) = \frac{1}{Q} \sum_{z=0}^{Q-1} P(A(z) = k). \quad (6)$$

Therefore, from (4) we have:

$$p_{ij}^Y = P(Y_{t+1} = j | Y_t = i) = \begin{cases} P(A = j), & \text{if } Y_t = 0, \\ P(A = j - i + 1), & \text{if } Y_t \geq 1. \end{cases} \quad (7)$$

Note that Eq. (4)-(7) correspond to the dynamics in Shortle et al. (2018, Section 6.1.2). Similarly to Shortle et al. (2018), the transition matrix of process Y_t , $t \geq 0$ can be written as:

$$P^Y = [p_{ij}^Y] = \begin{bmatrix} k_0 & k_1 & k_2 & \dots & k_M & 0 & 0 & \dots \\ k_0 & k_1 & k_2 & \dots & k_M & 0 & 0 & \dots \\ 0 & k_0 & k_1 & k_2 & \dots & k_M & 0 & \dots \\ 0 & 0 & k_0 & k_1 & k_2 & \dots & k_M & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}, \quad (8)$$

where M is the maximum number of new batch arrivals, i.e., the maximum value that random variable A can assume.

The steady state probabilities of process Y_t , $t \geq 0$ can then be recursively evaluated by solving:

$$\pi_Y^T = \pi_Y^T P, \quad (9)$$

with the contour condition that $\pi_Y(0) = 1 - \rho$ (Shortle et al., 2018, Eq. (6.12)), where $\rho = E(A)$. Observe that, to guarantee stability, we must have $E(A) < 1$. Note also that the steady state probabilities up to a given state j can be recursively calculated using the first j equations in (9).

4.3. Upper and lower bounds on the batch size Q under waiting time requirements

Let ω be the waiting time of a patient joining the queue, and assume that the healthcare system requires that:

$$P(\omega > \tau) < \alpha, \quad (10)$$

for some target waiting time $\tau > 0$ and probability threshold $\alpha \in (0, 1)$. Now, if a patient arrives and encounters $Y_t > 0$ batches of patients in the queue, their turn will only arrive once these Y_t batches are processed, which will take Y_t time periods. This means that the patient will have to wait $\tau = Y_t$ periods for their turn. Therefore, Eq. (10) can be equivalently stated as:

$$\sum_{i=0}^{\tau} \pi_Y(i) > 1 - \alpha, \quad (11)$$

where $\pi_Y : S_Y \rightarrow (0, 1)$ satisfies Eq. (9). Therefore, to limit the waiting time according to the prescribed requirements, we can establish a lower bound ($\underline{Q}$) on the batch size $Q > 0$ as follows:

$$\underline{Q} = \min\{Q \in \mathbb{N} : \sum_{i=0}^{\tau} \pi_Y(i) > 1 - \alpha\}. \quad (12)$$

When $Y_t = 0$, on the other hand, the next batch will only be processed when the system leaves this state, therefore by applying the strong Markov property (Brémaud, 2020), we have:

$$\omega = \min\{\delta > 0 : Y_\delta > 0 | Y_0 = 0\}.$$

Considering the transition matrix in Eq. (7), it follows that ω will be a geometrically distributed variable and:

$$P(\omega > \tau | Y_0 = 0) = k_0^\tau. \quad (13)$$

One can intuitively see that the value of k_0 will increase as Q increases, since in this case we will need more patient arrivals until we are able to process a full batch of patients. Hence, from Eq. (13), we can set up an upper bound $\bar{Q}$ for the batch size such that:

$$\bar{Q} = \max\{Q \in \mathbb{N} : k_0^\tau < \alpha\}. \quad (14)$$

In the example below, we show how to derive general bounds $\underline{Q}$ and $\bar{Q}$ for a given demand distribution and waiting time requirements.

Example 1 (Deriving lower and upper bounds for Q). *Consider an elective surgery queue with monthly demand D that satisfies the distribution in Table 3.*

Table 3: Monthly demand distribution for surgeries

Demand (d)	0	1	2	3	4	5	6	7	8	9	10
$P(D = d)$	0.0308	0.0393	0.1179	0.2096	0.2445	0.1956	0.1086	0.0414	0.0103	0.0015	0.0005

The expected monthly demand is therefore $\mathbb{E}(D) = 3.9022$. Setting $Q = 4$, which ensures long-term stability, we have $S_Z = \{0, 1, 2, 3\}$. This allows us to evaluate the individual distributions of $A(z)$, $z \in S_Z$ as defined in Eq. (5), which we depict in the Table 4, along with the distribution of variable A from Eq. (6).

The distributions in Table 4 follow directly from Eq. (5) and Eq. (6). For example, when $z = 0$, we know that the queue comprises an integer number of batches. Therefore, $P(A(0) = 0) = p_D(0) + p_D(1) + p_D(2) + p_D(3)$ (Table 3) as up to 3 new arrivals will not suffice to generate a new batch of $Q = 4$ patients. Analogously, $A(0) = 1$ if we observe from 5 to 7 arrivals and $A(0) = 2$ when 8 to 10 patients arrive in the period, as these will suffice for 2 new batches.

Conversely, when $z = 3$, we have three patients in excess of a discrete number of batches, which means that $P(A(3) = 0) = p_D(0)$ as new batches will be formed whenever one or more patients arrive. One new batch will

Table 4: Monthly demand distribution for surgeries

$P(A(z) = d)$	Batch demand (d)			
	0	1	2	3
$z = 0$	0.3976	0.5901	0.0123	0
$z = 1$	0.1880	0.7583	0.0537	0
$z = 2$	0.0701	0.7676	0.1618	0.0005
$z = 3$	0.0308	0.6113	0.3559	0.002
$P(A = d)$	0.171625	0.681825	0.145925	0.000625

form if $D \in [1, 4]$, two new batches if $D \in [5, 8]$ and three new batches when $D \in [8, 10]$. Specifically, when 10 patients arrive, they will form a set of 13 patients when joined to the $z = 3$ remaining patients from the start of the period. These 3 patients will then suffice to form three batches of $Q = 4$ patients, with one remaining batchless for the following period.

Note that $\mathbb{E}(A) = 0.9755$, and that this value is equal to $\rho = \frac{\mathbb{E}(D)}{Q} = \frac{3.9022}{4} = 0.9755$. This is expected, as the dynamics of process Y_t describes the original system, only looking at it in terms of batches of patients. Note that $\mathbb{E}(A)$ is simply the expected number of batches of Q patients arriving and since we process $\mu = 1$ batch per time period, we have $\rho = \mathbb{E}(A)$ when looking at the evolution of the batch process Y_t , $t \geq 0$.

Now, let us suppose that $\tau = 6$ and $\alpha = 0.05$ in (10), which means that the system requires 95% of the patients to be operated in up to six months. To verify whether that is satisfied, we need firstly to solve Eq. (9) for the steady state distribution of Y_t , $t \geq 0$ and then use it to verify (11). From Table 4, we obtain $k_0 = 0.171625$, $k_1 = 0.681825$, $k_2 = 0.145925$ and $k_3 = 0.000625$, which can be applied in Eq. (9) to find $\pi_Y(0) = 0.02445$, $\pi_Y(1) = 0.1182$, $\pi_Y(2) = 0.1219$, $\pi_Y(3) = 0.1046$, $\pi_Y(4) = 0.0898$, $\pi_Y(5) = 0.077$, $\pi_Y(6) = 0.0661$. This yields:

$$P(\omega \leq 6 | Y_t \geq 1) \approx 0.602 < 0.95 = 1 - \alpha.$$

This implies that $\underline{Q} > 4$ in Eq. (12) and we should, therefore, recalculate the probability above for increasing values of Q until we find the first value that satisfies the waiting time criterion - which will give us the lower bound $\underline{Q}$.

To verify the waiting time requirements when $Y_t = 0$, one can apply (13), which implies that $P(\omega \leq 6 | Y_t = 0) = 0.171625^6 \approx 2.5 \cdot 10^{-5} < 0.05$. Therefore, to find the bound $\underline{Q}$ in Eq. (14).

When verifying Eq. (12) with $Q = 10$, it implies $P(\omega \leq 6 | Y_t = 0) = 0.609780^6 \approx 0.05114 > 0.05$. Then, if one set $Q = 9$, it implies $P(\omega \leq 6 | Y_t = 0) = 0.566478^6 \approx 0.03304 < 0.05$. Thus, we can set $\underline{Q} = 9$.

For the $\underline{Q}$, one can verify $Q = 5$, which implies $k_0 = 0.265720$, $k_1 = 0.688120$, $k_2 = 0.046160$ and $k_3 = 0$, which can be applied in Eq. (9) to find $\pi_Y(0) = 0.21956$, $\pi_Y(1) = 0.606723$, $\pi_Y(2) = 0.143539$, $\pi_Y(3) = 0.024935$, $\pi_Y(4) = 0.004332$, $\pi_Y(5) = 0.000752$, $\pi_Y(6) = 0.000131$. This yields:

$$P(\omega \leq 6 | Y_t \geq 1) \approx 0.999973 > 0.95 = 1 - \alpha.$$

Thus, we can set $\underline{Q} = 5$.

5. Estimating the number of patients per time block

To ensure consistency with the waiting-time requirements established by the (R, Q) policy in Section 4, it is necessary to estimate how many elective patients can be accommodated within a single 6-hour operating theatre block, in accordance with the scheduling regulations at HC-Unicamp. This estimation determines whether the batch sizes $\underline{Q}$ and $\bar{Q}$, for each specialty $s \in \mathcal{S}$, defined at the strategic level can be effectively served within the planning horizon.

Accurate estimation requires explicit consideration of uncertainty in surgery durations, which varies across sub-specialties and directly affects both operating theatre utilisation and the risk of overtime. In particular, a trade-off arises between idle time, resulting from underutilised theatre capacity, and overtime, which may trigger cancellations of scheduled procedures.

For this purpose, in this section we present the second module of the proposed integrated framework. A newsvendor-based model is first employed to determine an optimal time allocation for the final surgery in a block under duration

uncertainty, balancing idle-time and overtime costs. Building on this result, an inventory-based formulation is then used to estimate the number of patients that can be reliably scheduled within a time block, providing the tactical input required for subsequent operating theatre allocation decisions.

5.1. Managing surgery cancellations via a newsvendor-based model

Let us now consider the problem of determining the optimal time budget for the last surgery in a session, based on the random duration of this surgery. The symbols used in this section are defined in Table 6.

Table 5: Symbols used in the newsvendor-based model.

Symbol	Description
Index	Description
s	Specialty index
Set	Description
$\mathcal{S}$	Set of specialties
Ω	Support of the surgery-duration random variable
Parameter	Description
T	Length of an operating theatre block
t_b	Time budget allocated to the last surgery in a block
c	Unit cost of unused allocated time
p	Unit penalty cost of overtime
t^*	Safety time threshold for starting the next surgery
Random Variable	Description
T_D	Random duration of a surgery
β_1	Unused allocated time
β_2	Overtime duration
Function	Description
$f_{T_D}(\cdot)$	Probability mass function of T_D
$F_{T_D}(\cdot)$	Cumulative distribution function of T_D
$u(t_b)$	Expected cost of allocating time budget t_b

Let T_D be a non-negative random variable taking values from $\Omega \in \mathbb{R}_+$, which represents the duration of a given surgery, with cumulative distribution $F_{T_D} : \Omega \rightarrow [0, 1]$. Assume that, once the theatre is ready for the surgery, it only proceeds as planned if the remaining time in the session exceeds a prescribed budget of $t_b \in \Omega$ time units. Otherwise, the surgery is called off.

The time budget $t_b \in \Omega$ can be interpreted as a perishable inventory of session time. If the inventory is excessive, we will not fully utilise the time allocated to the surgical team to perform the surgery. Conversely, if we proceed with the surgery, but the time budget is insufficient, the surgical team will have to work overtime to complete the surgery. Let $c > 0$ be the cost of each unit of time budget in excess of the actual surgical time T_D and $p > c$ be a penalty for each unit of overtime that the surgical team incurs. The expected cost of the time deviation between T_D and t_b is defined as:

$$u(t_b) = c\mathbb{E}(\beta_1) + p\mathbb{E}(\beta_2), \quad (15)$$

where $\beta_1 = \min\{t_b - T_D, 0\}$ is the underused time budget, and $\beta_2 = \max\{T_D - t_b, 0\}$ is the total overtime. The objective is to find an optimal time budget allocation t^* that minimizes the time deviation cost in Eq. (15), which is given by:

$$t^* = \arg \min_{t_b \in \Omega} u(t_b). \quad (16)$$

Lemma 1. Let $\beta_1 = \min\{t_b - T_D, 0\}$ and $\beta_2 = \max\{T_D - t_b, 0\}$. Then, it follows that:

$$\mathbb{E}(\beta_1) = \int_0^{t_b} F_{T_D}(t)dt, \quad \mathbb{E}(\beta_2) = \int_{t_b}^{\infty} (1 - F_{T_D}(t))dt \quad (17)$$

Proof. We will start by proving the result for β_1 . Since $\beta_1 \geq 0$, it follows that:

$$\mathbb{E}(\beta_1) = \int_{t=0}^{\infty} P(\beta_1 > t)dt = \int_{t=0}^{t_b} P(t_b - T_D > t)dt = \int_{t=0}^{t_b} P(T_D < t_b - t)dt = \int_{t=0}^{t_b} F_{T_D}(t_b - t)dt = \int_{t=0}^{t_b} F_{T_D}(t)dt.$$

Since β_2 is also non-negative, we can write:

$$\mathbb{E}(\beta_2) = \int_0^\infty P(T_D - t_b > t)dt = \int_0^\infty P(T_D > t + t_b)dt = \int_{t_b}^\infty P(T_D > t)dt = \int_{t_b}^\infty (1 - F_{T_D}(t))dt$$

□

Theorem 1. Let $u(t_b)$ be defined in Eq. (15), and t^* be a solution to Eq. (16). Then, it follows that:

$$t^* = F_{T_D}^{-1}\left(\frac{\frac{p}{c}}{1 + \frac{p}{c}}\right), \quad (18)$$

where $F_{T_D}^{-1}(\cdot)$ is the inverse of the cumulative distribution function $F_{T_D}(\cdot)$.

Proof. Substituting (17) in Eq. (15) yields:

$$u(t_b) = c \int_0^{t_b} F_{T_D}(t)dt + p \int_{t_b}^\infty (1 - F_{T_D}(t))dt.$$

When $t_b = t^*$, the first order optimality condition for $u(t_b)$ implies:

$$\frac{du(t_b)}{dt_b} = cF_{T_D}(t^*) - p(1 - F_{T_D}(t^*)) = 0.$$

Hence, we must have:

$$cF_{T_D}(t^*) - p(1 - F_{T_D}(t^*)) \implies \frac{F_{T_D}(t^*)}{(1 - F_{T_D}(t^*))} = \frac{p}{c} \implies F_{T_D}(t^*)\left(1 + \frac{p}{c}\right) = \frac{p}{c}$$

We conclude the proof by noting that (18) follows from the expression above. □

Theorem 1 implies that, whenever the remaining time in a session exceeds the quantile t^* in Eq. (18), the surgery is greenlighted to continue. Otherwise, the surgery is cancelled and the session comes to a close. Note that the quantile is solely a function of the ratio between the overtime cost p and the cost of unused session time c , and is applicable to any surgery time distribution. For example, if $p = 2c$, $t^* = F_{T_D}^{-1}\left(\frac{2}{3}\right)$. This means that at least two thirds of the green-lighted surgeries will not require overtime.

5.2. Applying the newsvendor model to the case study

Building on the theoretical results of the previous subsection, we first estimate, for each surgical subspecialty, the distribution of the number of surgeries that can be completed within a block of length $T \in \mathbb{Z}_+$. This distribution is then used in the subsequent subsection to evaluate alternative planning decisions under an inventory-based formulation.

We use the notation introduced in Table 6. For each subspecialty, let d_i denote the non-negative random variable representing the duration of the i -th surgery, with probability mass function f_d and cumulative distribution function F_d . We assume that surgery durations are independent and identically distributed within each subspecialty. Hence, for any $k \in \mathbb{N}_+$, the cumulative duration of the first k surgeries is

$$cd_k = \sum_{i=1}^k d_i.$$

Let f_{cd_k} and F_{cd_k} denote the probability mass function and cumulative distribution function of cd_k , respectively.

After k surgeries have been completed, one further surgery can be accommodated only if the residual time in the block exceeds the safety threshold t^* . herefore,

$$P(K > k) = P(T - cd_k > t^*) = P(cd_k < T - t^*). \quad (19)$$

Table 6: Symbols used in the derivation of the completion distribution.

Set / Index / Parameter	Description
$\mathcal{S}$	Set of surgical subspecialties
s	Index of a surgical subspecialty, $s \in \mathcal{S}$
T	Length of the operating block
t^*	Safety time threshold that must remain available in the block to accommodate one further surgery
$\varepsilon_{\text{stop}}$	Probability threshold used to truncate the tail of the distribution
K_{max}	Largest value of k such that $P(K > k) > \varepsilon_{\text{stop}}$
Random Variable	Description
d_i	Duration of the i -th surgery for a given subspecialty
cd_k	Cumulative duration of the first k surgeries, $cd_k = \sum_{i=1}^k d_i$
K	Number of surgeries that can be completed within a block
Function / Derived Quantity	Description
f_d	Probability mass function of the duration of a single surgery
F_d	Cumulative distribution function of the duration of a single surgery
f_{cd_k}	Probability mass function of cd_k
F_{cd_k}	Cumulative distribution function of cd_k
f_K	Probability mass function of K
F_K	Cumulative distribution function of K
$*$	Convolution operator
q	Vector with entries $q[k] = P(K > k)$

Thus, the event $\{K > k\}$ is equivalent to the event that, after completing k surgeries, sufficient time remains in the block to start one additional surgery while preserving the required safety buffer.

Because cd_k is the sum of k independent surgery durations, its distribution can be obtained recursively by convolution:

$$f_{cd_1} = f_d, \quad f_{cd_k} = f_{cd_{k-1}} * f_d, \quad k \geq 2.$$

For each k , Eq. (19) then yields the survival probability $P(K > k)$.

These survival probabilities are computed sequentially and stored in the vector q , whose k -th entry is defined as

$$q[k] = P(K > k).$$

The procedure starts at $k = 0$, with $q[0] = 1$, and stops when $q[k] \leq \varepsilon_{\text{stop}}$. This truncation preserves the relevant support of the distribution while avoiding unnecessary convolution steps in the negligible tail. For each subspecialty, the algorithm therefore returns both the vector q and the largest relevant index K_{max} .

Algorithm 1 Computation of $P(K > k)$ via convolutions

```

1: Input:  $T, t^*, \varepsilon_{\text{stop}}$ , subspecialties  $s \in \mathcal{S}$ , and duration distributions  $f_d$  for each  $s \in \mathcal{S}$ 
2: Output: vector  $q$  such that  $q[k] = P(K > k)$ 
3: for each  $s \in \mathcal{S}$  do
4:   Initialise  $q[0] \leftarrow 1$ 
5:   Initialise  $k \leftarrow 1$ 
6:   Initialise  $f_{cd_1} \leftarrow f_d$ 
7:   while  $q[k-1] > \varepsilon_{\text{stop}}$  do
8:     if  $k > 1$  then
9:        $f_{cd_k} \leftarrow f_{cd_{k-1}} * f_d$ 
10:    end if
11:     $q[k] \leftarrow P(cd_k < T - t^*)$ 
12:     $k \leftarrow k + 1$ 
13:  end while
14:   $K_{\text{max}} \leftarrow k - 1$ 
15: end for
```

Once the survival function has been obtained, the probability mass function of K follows from the standard relationship between the PMF and the survival function of a discrete random variable:

$$P(K = k) = P(K > k - 1) - P(K > k), \quad k \geq 1. \quad (20)$$

In addition,

$$P(K = 0) = 1 - P(K > 0).$$

Hence, the convolution procedure yields the full probability distribution of the number of surgeries that can be completed within the block for each subspecialty. This distribution is the main output of the present subsection and constitutes the probabilistic input to the inventory-based planning model described next.

5.3. Inventory-based model for defining the number of surgeries in a time block

TODO: ADD CITATIONS ABOUT INVENTORY MODEL USED FOR MODELING HERE

Using the distribution of K derived in the previous subsection, we now determine the number of surgeries to be planned within a single operating theatre block. This decision is formulated through an inventory-based perspective that balances the marginal benefit of scheduling additional surgeries against the expected cost of cancellations. The symbols used in this subsection are defined in Table 7.

Table 7: Symbols used in the inventory-based planning model.

Parameter / Decision Variable	Description
n	Number of surgeries planned in a block
n^*	Optimal number of surgeries planned in a block
ℓ	Number of cancelled surgeries
g	Marginal value associated with scheduling one additional surgery
Function / Derived Quantity	Description
$G(n)$	Marginal gain associated with planning n surgeries
$cost(\ell)$	Cost incurred when ℓ surgeries are cancelled
$C(n)$	Expected cancellation cost when n surgeries are planned
$\Delta(n)$	Net value of planning n surgeries
$P(\text{cancellations} = \ell \mid n)$	Probability of cancelling ℓ surgeries when n surgeries are planned

Let $n \in \mathbb{N}_+$ denote the number of surgeries planned in a block. If the random number of surgeries that can actually be completed is K , then the number of cancellations is given by

$$\ell = (n - K)^+,$$

where $(x)^+ = \max\{x, 0\}$. Hence, for $\ell = 0, 1, \dots, n$,

$$P(\text{cancellations} = \ell) = P(K = n - \ell), \quad \ell = 1, \dots, n, \quad (21)$$

and

$$P(\text{cancellations} = 0) = P(K \geq n).$$

We next associate each planning decision with a marginal gain and an expected cancellation cost. In contrast to classical inventory formulations, in which the reward is usually proportional to the expected number of units served, we define the gain of planning n surgeries as the value of successfully accommodating the n -th scheduled surgery:

$$G(n) = g \cdot P(K \geq n), \quad (22)$$

where $g > 0$ is a constant representing the marginal value of scheduling one additional surgery. Since

$$P(K \geq n) = P(K > n - 1),$$

Eq. (22) is directly obtained from the survival probabilities computed in the previous subsection.

In this setting, $G(n)$ is not interpreted as a direct monetary return, but rather as the marginal benefit associated with increasing the number of planned surgeries from $n - 1$ to n . This interpretation is particularly suitable in public healthcare settings, where the value of additional surgeries may reflect improved access to care, reduced waiting times, and better utilisation of operating theatre capacity.

The cost associated with cancellations is modelled through a function $cost : \mathbb{N}_+ \cup \{0\} \rightarrow \mathbb{R}_+$ satisfying $cost(0) = 0$ and assumed to be non-decreasing. The expected cancellation cost under planning decision n is therefore

$$C(n) = \mathbb{E}[cost((n - K)^+)] = \sum_{x=0}^{\infty} P(K = x) cost((n - x)^+). \quad (23)$$

The net value associated with planning n surgeries is then defined as

$$\Delta(n) = G(n) - C(n). \quad (24)$$

Because $G(n)$ represents the marginal value of adding the n -th surgery, the optimal planning level is the largest value for which the marginal net value remains positive:

$$n^* = \max\{n \in \mathbb{N}_+ : \Delta(n) > 0\}. \quad (25)$$

Thus, n^* identifies the point at which the expected benefit of adding one more planned surgery is no longer sufficient to offset the corresponding expected cancellation cost.

Once n^* has been determined, the relevant probabilistic input for the subsequent operating theatre allocation model is the truncated distribution of the number of surgeries completed under that planning decision, i.e.,

$$P(K = k), \quad k = 0, 1, \dots, n^*,$$

together with the implied probability of cancellation. These quantities are then passed to the next stage of the framework.

6. Operating Theatre Scheduling Problem

Following the proposed integrated framework, the final step consists of solving the Operating Theatre Scheduling Problem (OTSP), which addresses the tactical allocation of subspecialties to operating theatre blocks. This stage is essential for translating the strategic decisions derived from the preceding layers into actionable schedules that can be implemented in hospital operations. To this end, the model takes as input the demand for each subspecialty, driven by the (R, Q) policy, as well as the effective capacity of each block, determined in the cancellation and overtime management stage.

We formulate the OTSP as a Mixed-Integer Linear Programming (MILP) model, which enables the representation of complex constraints and multiple, potentially competing objectives, while ensuring optimality with respect to the specified criteria. The model incorporates several objectives aligned with hospital priorities, including maximising operating theatre utilisation, ensuring the allocation of priority subspecialties, and promoting efficient sequencing of block allocations from a logistical perspective.

Nevertheless, the proposed framework is sufficiently flexible to accommodate alternative solution approaches, such as heuristics or metaheuristics. This flexibility arises because the outputs of the previous layers, namely demand and effective capacity, can serve as inputs to any scheduling methodology, not exclusively the MILP formulation proposed here. This highlights the value of integrating the different decision layers, as it provides a structured mechanism for incorporating both strategic and operational insights into the scheduling process, irrespective of the specific solution approach adopted for the OTSP.

In the following subsections, we first describe the preprocessing of strategic inputs derived from the (R, Q) policy and the effective capacity information in order to determine subspecialty demand in terms of block units, which constitute the relevant unit for the scheduling problem. We then present the MILP formulation of the OTSP, detailing the decision variables, objective functions, and constraints that reflect the hospital's operational requirements and priorities.

6.1. Preprocessing Strategic Input

The OTSP requires demand to be expressed in block units, as these correspond to the operational granularity of theatre scheduling. To this end, we preprocess the strategic outputs of the (R, Q) policy together with the effective capacity information obtained from the cancellation and overtime management stage.

Recall that the overtime and cancellation module provides the probability mass function of planning of the optimal number of surgeries to be planned within a single block (i.e., $P(K = k)$ for $k \in 1, \dots, n^*$) for subspecialty s . To determine the block requirements associated with targets $\underline{Q}$ and $\bar{Q}$ at confidence level α , we iteratively convolve this distribution.

Let $F_s^{(n)}$ denote the cumulative distribution function associated with the PMF obtained after convolving $block_pmf_s$ with itself n times. Then, for a generic target $Q' \in \{\underline{Q}, \bar{Q}\}$, the minimum number of blocks required is the smallest integer n such that

$$1 - F_s^{(n)}[Q' - 1] \geq \alpha.$$

In other words, we seek the smallest number of blocks n such that the probability of completing at least Q' surgeries is at least α . Algorithm 2 summarises this procedure.

Algorithm 2 Block demand estimation via successive convolutions

```

1: Input: set of subspecialties  $\mathcal{S}$ , targets  $(\underline{Q}, \bar{Q})$ , confidence level  $\alpha$ , PMF  $block\_pmf_s$  of surgeries per block for each  $s \in \mathcal{S}$ 
2: Output: block requirements  $(B_s, \bar{B}_s)$  for each  $s \in \mathcal{S}$ 
3: for each subspecialty  $s \in \mathcal{S}$  do
4:   if  $\underline{Q} = 0$  and  $\bar{Q} = 0$  then
5:      $B_s \leftarrow 0, \bar{B}_s \leftarrow 0$ 
6:   else
7:     for each  $Q' \in \{\underline{Q}, \bar{Q}\}$  do
8:        $counter \leftarrow 1$ 
9:        $block\_pmf_s^{(counter)} \leftarrow block\_pmf_s$ 
10:       $F \leftarrow \text{CalculateCDF}(block\_pmf_s^{(counter)})$ 
11:      while  $1 - F[Q' - 1] < \alpha$  do
12:         $block\_pmf_s^{(counter+1)} \leftarrow \text{Convolve}(block\_pmf_s^{(counter)}, block\_pmf_s)$ 
13:         $counter \leftarrow counter + 1$ 
14:         $F \leftarrow \text{CalculateCDF}(block\_pmf_s^{(counter)})$ 
15:      end while
16:      if  $Q' = \underline{Q}$  then
17:         $B_s \leftarrow counter$ 
18:      else
19:         $\bar{B}_s \leftarrow counter$ 
20:      end if
21:    end for
22:  end if
23: end for

```

This step is essential for the OTSP phase of the framework, since the problem is formulated at the block level and the monthly operating theatre schedule is constructed by allocating blocks rather than individual surgeries. Moreover, this preprocessing step enables the uncertainty captured in the strategic layer to be incorporated into the framework without introducing stochastic elements directly into the scheduling model. This simplifies both the model formulation and the solution process (i.e., the OTSP is solved as a deterministic MILP), while ensuring that the resulting schedules remain aligned with demand targets that reflect both variability in surgical durations and the effective capacity of blocks.

6.2. MILP Formulation

We model the OTSP as a MILP, which allows us to capture the complex interactions between subspecialty demand, block capacity, and scheduling constraints in a structured manner. To formulate the problem, we use the notation in Table 8.

The Objective Functions (OFs) are organised according to the priority levels established by the hospital's guidelines – although they can be reordered as needed to reflect different institutional priorities in other settings. To reflect these priorities, we adopt a hierarchical optimisation strategy, as proposed by Aringhieri et al. (2022). The OFs are optimised sequentially, while respecting the optimal values obtained in the previous stages.

Formally, the first OF is defined as

$$\text{Min } F_0 = \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} z_{sb} + \gamma \cdot a_b, \quad (26)$$

which minimises the total slack in team and anesthesiologist availability, with a high penalty on the latter as they are more scarce and expensive. The second function is given by

$$\text{Min } F_1 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{otbs} \text{ only for variables where } \lambda_{otbs} = 0 \quad (27)$$

Table 8: Table of Symbols

Set	Description
OT	Operating theatres (both elective and outpatient)
$\mathcal{D}$	Days of the planning horizon, such that $\mathcal{D} = \{1, 2, \dots, \mathcal{D} \}$
$\mathcal{M}$	Periods (morning or afternoon), such that $\mathcal{M} = \{\text{morning}, \text{afternoon}\}$
$\mathcal{B}$	Blocks, such that $\mathcal{B} = \mathcal{D} \times \mathcal{M}$
$\mathcal{S}$	Surgical subspecialties
$\mathcal{S}_{seq}$	Surgical subspecialties that necessarily require two sequential allocated blocks, such that $\mathcal{S}_{seq} \subset \mathcal{S}$
$\mathcal{S}_{prio}$	but have high priority, such that $\mathcal{S}_{circ} \subset \mathcal{S}$
$\mathcal{S}_{tx}$	Priority surgical subspecialties, such that $\mathcal{S}_{prio} \subset \mathcal{S}$
$\mathcal{S}_{tx}$	Organ transplant surgical subspecialties, such that $\mathcal{S}_{tx} \subset \mathcal{S}$
Index	Description
ot	Operating theatre, with $ot \in OT$
d	Planning horizon day, with $d \in \mathcal{D}$
m	Period (morning or afternoon), with $m \in \mathcal{M}$
b	Surgery block, with $b \in \mathcal{B}$
s	Surgical subspecialty, with $s \in \mathcal{S}$
Parameter	Description
$\bar{B}_s$	Maximum number of blocks allocated to each subspecialty $s \in \mathcal{S}$
B_s	Minimum number of blocks allocated to each subspecialty $s \in \mathcal{S}$
$NumTeams_{bs}$	Number of teams of subspecialty $s \in \mathcal{S}$ available in block $b \in \mathcal{B}$
$NumAnest_{bs}$	Number of anesthesiologists available in block $b \in \mathcal{B}$
$NeedAnest_s$	1 if subspecialty $s \in \mathcal{S}$ requires an anesthesiologist; 0 otherwise
γ	Constant to penalize the anesthesiologist slack variable a_b in the objective function ($\gamma = \mathcal{B} \cdot \mathcal{S} \cdot OT $)
λ_{otbs}	1 if room $ot \in OT$, in block $b \in \mathcal{B}$, can be used by subspecialty $s \in \mathcal{S}$ in the shift block $b \in \mathcal{B}$; 0 otherwise
Variable	Description
x_{otbs}	1 if room $ot \in OT$, in block $b \in \mathcal{B}$, is allocated to subspecialty $s \in \mathcal{S}$; 0 otherwise.
y_{otds}	This variable is only created if the allocation is feasible (i.e., specialty s is compatible with the theatre ot).
w_{bs}	1 when room $ot \in OT$, on day $d \in \mathcal{D}$, is allocated only to subspecialty $s \in \mathcal{S}$; 0 otherwise
a_b	Number of transplants to be performed in block $b \in \mathcal{B}$ by subspecialty $s \in \mathcal{S}$
z_{sb}	Slack variable for the number of anesthesiologists available in block $b \in \mathcal{B}$
z_{sb}	Slack variable for the number of teams available in block $b \in \mathcal{B}$ for subspecialty $s \in \mathcal{S}$

which minimises the number of allocations that violate the usual availability of rooms for each subspecialty, thus promoting the use of preferred rooms and reducing the need for last-minute adjustments. The third objective function is defined as

$$\text{Max } F_2 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}_{prio}} x_{otbs}, \quad (28)$$

which maximises the allocation of priority subspecialties, ensuring that they receive as much capacity as possible within the constraints of the system. The fourth OF is given by

$$\text{Max } F_3 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{otbs}, \quad (29)$$

which maximises the total number of allocated blocks, thus promoting overall capacity utilisation and service provision. Finally, the fifth OF is defined as

$$\text{Max } F_4 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{otbs} \quad (30)$$

which maximises the number of sequentially allocated blocks. This is particularly important for improving the logistics of the surgical process, reducing the need for equipment and team transfers between theatres.

With this, we formulate the theatre planning problem at HC - Unicamp with the model OTScheduling, incorporating these OFs and constraints based on the hospital's current operational framework, but with the flexibility to accommodate other settings as needed.

(*OTScheduling*) Optimize F_0-F_5

subject to

$$\sum_{ot \in OT} \sum_{b \in \mathcal{B}} x_{otbs} \leq \bar{B}_s \quad \forall s \in \mathcal{S} \quad (31)$$

$$\sum_{ot \in OT} \sum_{b \in \mathcal{B}} x_{otbs} \geq \underline{B}_s \quad \forall s \in \mathcal{S} \quad (32)$$

$$\sum_{s \in \mathcal{S}} x_{otbs} \leq 1 \quad \forall ot \in OT, b \in \mathcal{B} \quad (33)$$

$$\sum_{ot \in OT} \sum_{s \in \mathcal{S}} NeedAnest_s \cdot x_{otbs} \leq NumAnests_b + a_b \quad \forall b \in \mathcal{B} \quad (34)$$

$$\sum_{ot \in OT} x_{otbs} \leq NumTeams_{bs} + z_{sb} \quad \forall b \in \mathcal{B}, s \in \mathcal{S} \quad (35)$$

$$x_{otbs} + x_{ot(b+1)s} - 1 \leq y_{otds} \quad \forall ot \in OT, b \in \mathcal{B}, s \in \mathcal{S}, \\ d \in \mathcal{D} : b \text{ and } b+1 \text{ are blocks} \\ \text{of the same day } d \quad (36)$$

$$x_{otbs} + x_{ot(b+1)s} \geq 2 \cdot y_{otds} \quad \forall ot \in OT, b \in \mathcal{B}, s \in \mathcal{S}, d \in \mathcal{D} : \\ d \in \mathcal{D} : b \text{ and } b+1 \text{ are blocks} \\ \text{of the same day } d \quad (37)$$

$$x_{otbs} = x_{ot(b+1)s} \quad \forall ot \in OT, b \in \mathcal{B}', s \in \mathcal{S}_{seq}, \\ d \in \mathcal{D} : b \text{ and } b+1 \text{ are blocks} \\ \text{of the same day } d \quad (38)$$

$$\sum_{ot \in OT} x_{otbs} = 2 \cdot w_{bs} \quad \forall b \in \mathcal{B}, s \in \mathcal{S}_{tx} \quad (39)$$

$$x_{otbs} \in \{0, 1\} \quad \forall ot \in OT, b \in \mathcal{B}, s \in \mathcal{S}$$

$$y_{otds} \in \{0, 1\} \quad \forall ot \in OT, d \in \mathcal{D}, s \in \mathcal{S}$$

$$w_{bs} \in \mathbb{Z}_+ \quad \forall b \in \mathcal{B}, s \in \mathcal{S}$$

$$a_b \in \mathbb{Z}_+ \quad \forall b \in \mathcal{B}$$

$$z_{sb} \in \mathbb{Z}_+ \quad \forall b \in \mathcal{B}, s \in \mathcal{S}$$

Constraints (31) and (32) impose minimum and maximum bounds on the number of allocations of each subspecialty. This is part of the integration with the strategic layer, as these bounds are determined based on the demand surgeries demand for each subspecialty, which in turn is influenced by the (R, Q) policy. Constraint (33) ensures that each theatre and time block receives at most one subspecialty. Constraint (34) is a soft constraint that limits the number of allocations that require anaesthetists according to their availability $NumAnests_b$ in each block, while allowing for some flexibility through the slack variable a_b . Constraint (35) similarly limits the number of allocations for each subspecialty based on the availability of medical teams $NumTeams_{bs}$, with the slack variable z_{sb} providing flexibility when necessary. Constraints (36) and (37) together guarantee sequential allocation of blocks on the same day when $y_{otds} = 1$. Subspecialties requiring two consecutive blocks are specifically addressed by constraint (38). Constraint (39) ensures that transplant subspecialties are allocated in pairs, reflecting the need for simultaneous operating theatre use for donor and recipient procedures.

7. Computational Study

We conduct a computational study to evaluate the performance of the proposed approach using real-world data from HC - Unicamp. While the theoretical results provide guarantees on system stability, it is essential to assess

the practical implications in a realistic setting. To this end, we simulate the long-term surgery planning process and compare the proposed approach with a baseline policy in terms of patient waiting times and resource utilisation.

7.1. Real-world Data

The dataset comprises detailed information on surgery durations, operating theatre availability, team constraints, and specialty schedules over a one-year period (2025). Each month is treated as an individual instance of the operating theatre scheduling problem, while the full year is used to evaluate long-term system behaviour.

7.2. Derivation of Model Inputs

The data are processed to derive the inputs required for the strategic and operational components of the proposed approach.

Surgery duration distributions. For each subspecialty, empirical distributions of surgery durations are estimated from historical data. These distributions are used in the Overtime and Cancellation module to evaluate the newsvendor model and determine the number of surgeries scheduled per block.

Demand distributions. Patient arrivals are modelled as Poisson processes. For each subspecialty, the arrival rate λ is estimated as the average monthly demand derived from historical activity. This is obtained by combining the expected number of surgeries per operating block (estimated from the duration distributions) with the observed number of allocated blocks, yielding an average monthly arrival rate.

Operational parameters. All parameters of the operating theatre scheduling model (see Table 8 in Section 6) are calibrated using real-world data, i.e., the actual values observed in each month of the year 2025. These include operating theatre compatibilities, staff availability, and working calendars.

7.3. Experimental Setup

Experiments are conducted on a machine equipped with an Intel Core 7 150U processor (5.40 GHz) and 32 GB of RAM. All modules are implemented in Python 3.12, and the MILP model is solved using SCIP 9.2.3 with default parameter settings, interfaced through the Pyomo library.

For the Queue Management module, the waiting time target is set to $\tau = 6$ months with confidence level $\alpha = 0.1$, ensuring that patients are treated within this horizon with high probability. The threshold R is defined as the midpoint between $\underline{Q}$ and $\bar{Q}$.

In the Overtime and Cancellation module, the overtime cost is set to twice the regular surgery cost, while the cancellation cost equals the regular cost. Additionally, we set the cost function to be quadratic in the number of surgeries cancelled, i.e., $c(x) = x^2$, reflecting the increasing marginal cost of cancellations as more surgeries are cancelled.

These strategic parameters are computed once and remain fixed throughout the simulation horizon.

Baseline policy. The baseline policy excludes the strategic modules. All specialties are scheduled in every period, and the MILP model is solved using only operational constraints, without (R, Q) control or overtime optimisation.

7.4. Simulation Framework

Figure 4 illustrates the simulation flow. The system evolves over a monthly decision cycle. In each period, patient arrivals are realised according to the estimated demand distributions, and queues are updated accordingly. Under the proposed approach, only specialties with queue size exceeding R are selected for scheduling.

The MILP model then determines the allocation of operating theatre blocks. For the proposed approach, the number of treated patients is sampled based on the distributions obtained from the Overtime and Cancellation module. For the baseline policy, the number of treated patients is determined directly from surgery duration distributions without adjustment for overtime or cancellations. Queues are updated after treatment, and the process repeats for the next period.

To assess long-term stability, we perform 100 independent simulation runs using different random seeds to capture the stochasticity in surgery durations and patient arrivals.

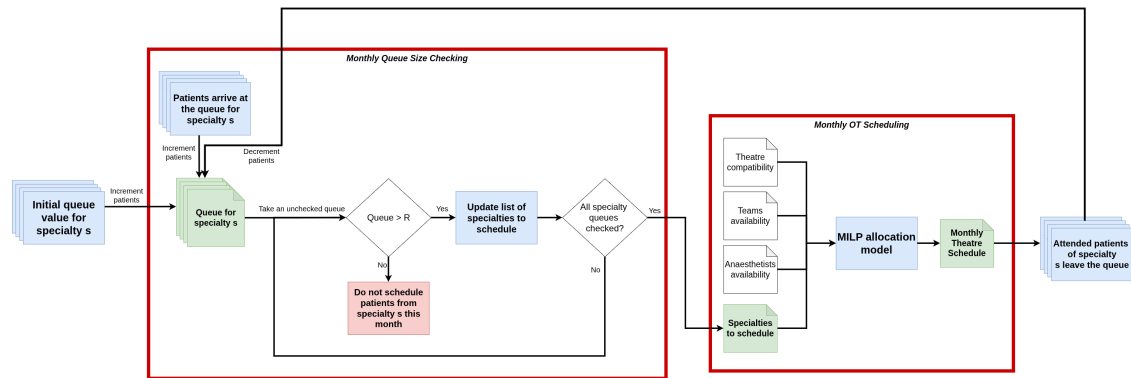


Figure 4: Long-term simulation flow.

7.5. Results from Strategic Decision-Making

The strategic modules are designed to ensure long-term stability and manage patient queues effectively. Together, the (R, Q) control and the Overtime and Cancellation module aim to balance demand and capacity, preventing excessive queue growth while optimising resource utilisation. The results from this decision level are analysed in two parts: the (R, Q) control outcomes and the impact of overtime and cancellations on system performance.

7.5.1. (R, Q) Results

7.5.2. Overtime and Cancellation Results

7.6. Long-term Stability Analysis

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